

Introduction

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PhD Student in Food Science working on epidemiology, in Dr. Renata Ivanek's lab

BS in Computer Science and Crop Sciences from the University of Illinois



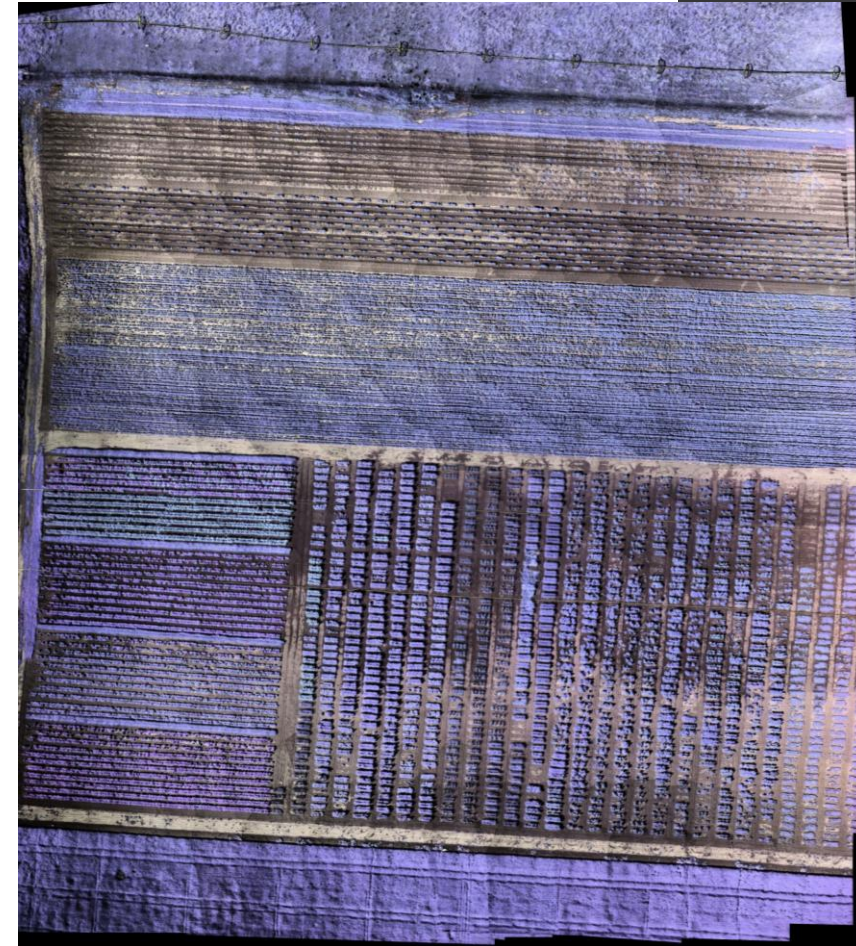
Work Adventures

Working in:

- Mastitis Prediction and Decision Tools for Dairy Farmers
- Data Privacy for Food Safety

Worked as:

- Data Engineer
- Plant Breeder (16 different crops)
- Farm Consultant in Morocco



Dimension Reduction

Dress for Success (Choosing what to wear)

Cousin's closet

- If they had 1 outfit in their closet...



Dress for Success (Choosing what to wear)

Cousin's closet

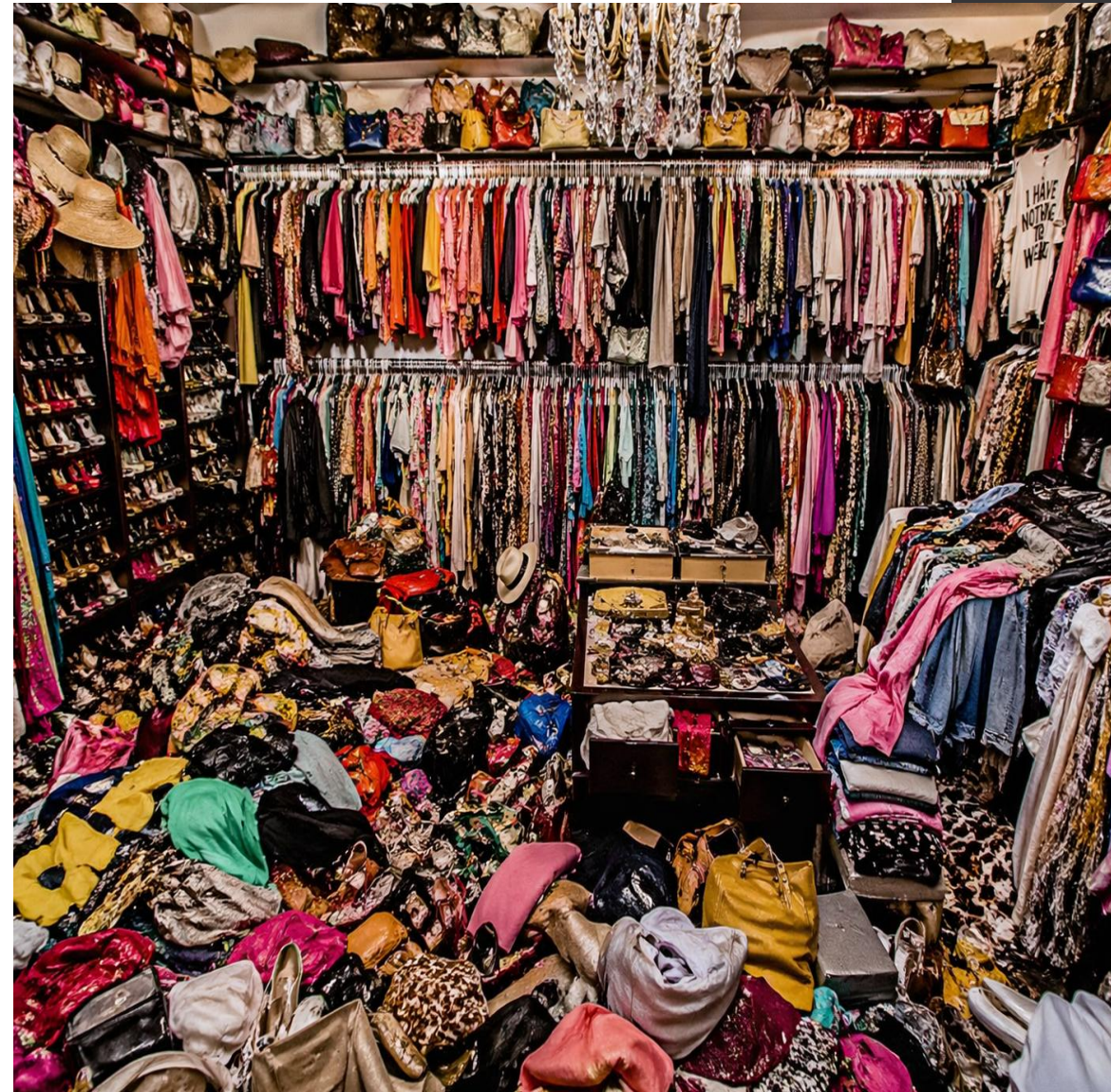
- If they had 50 outfits in their closet...



Dress for Success (Choosing what to wear)

Cousin's closet

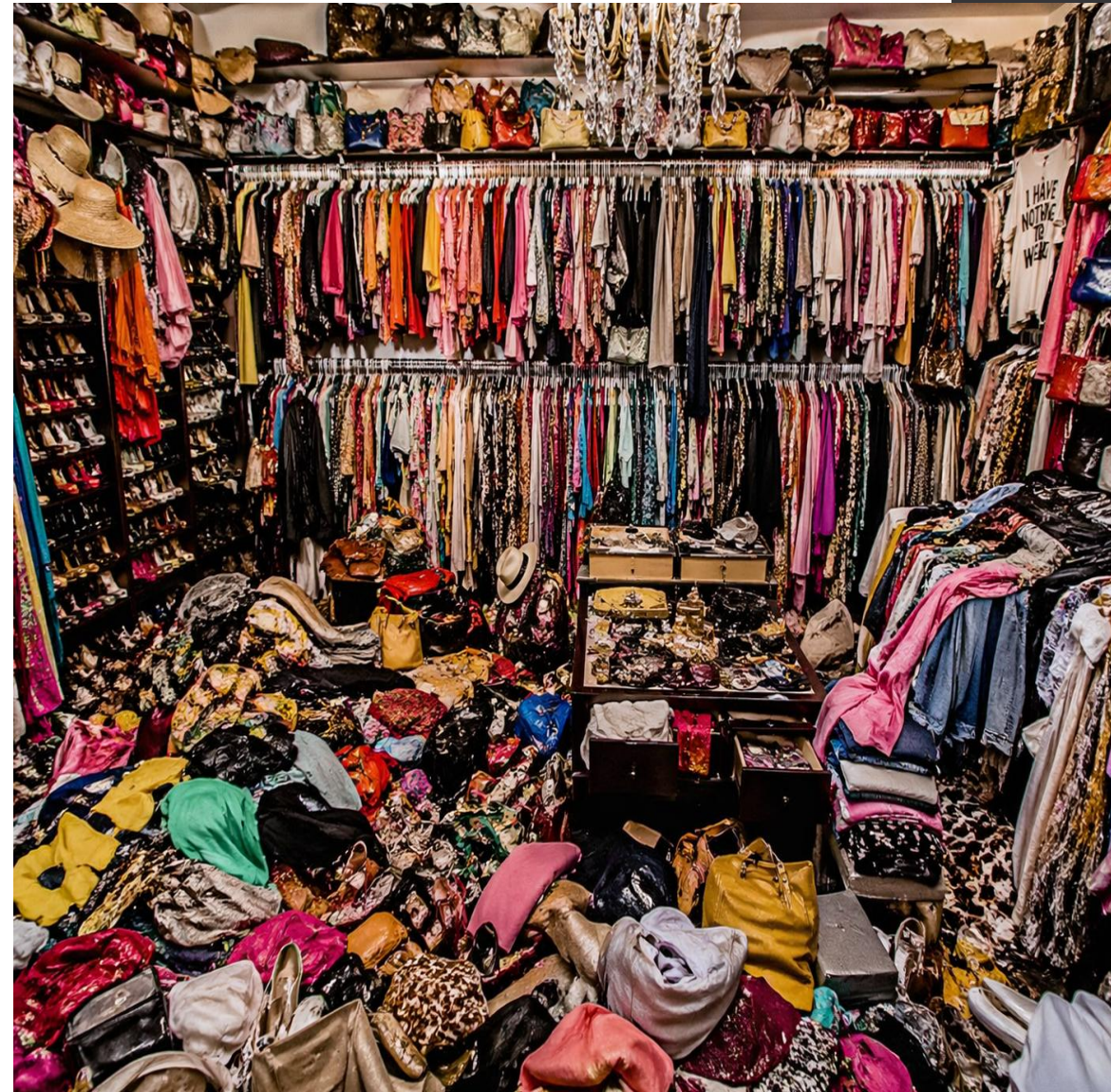
- If they had 1,000,000 outfits in their closet...



Dress for Success (Choosing what to wear)

Cousin's closet

- If they had 1,000,000 outfits in their closet...
- How can you make your decision easier?



Mastitis (udder inflammation) Prediction from Milk Data

How many factors go into
predicting Mastitis?



Mastitis (udder inflammation) Prediction from Milk Data

How many factors go into
predicting Mastitis?

How easy is it to predict
Mastitis in dairy cows?



Canine Kidney Disease from Blood Work

How many factors go into predicting Kidney Disease from bloodwork?



Canine Kidney Disease from Blood Work

How many factors go into predicting Kidney Disease from bloodwork?

How easy is it to predict Kidney disease in dogs?



Agenda

- What is Dimension Reduction
 - Why is it important?
 - When should we use it?



Agenda

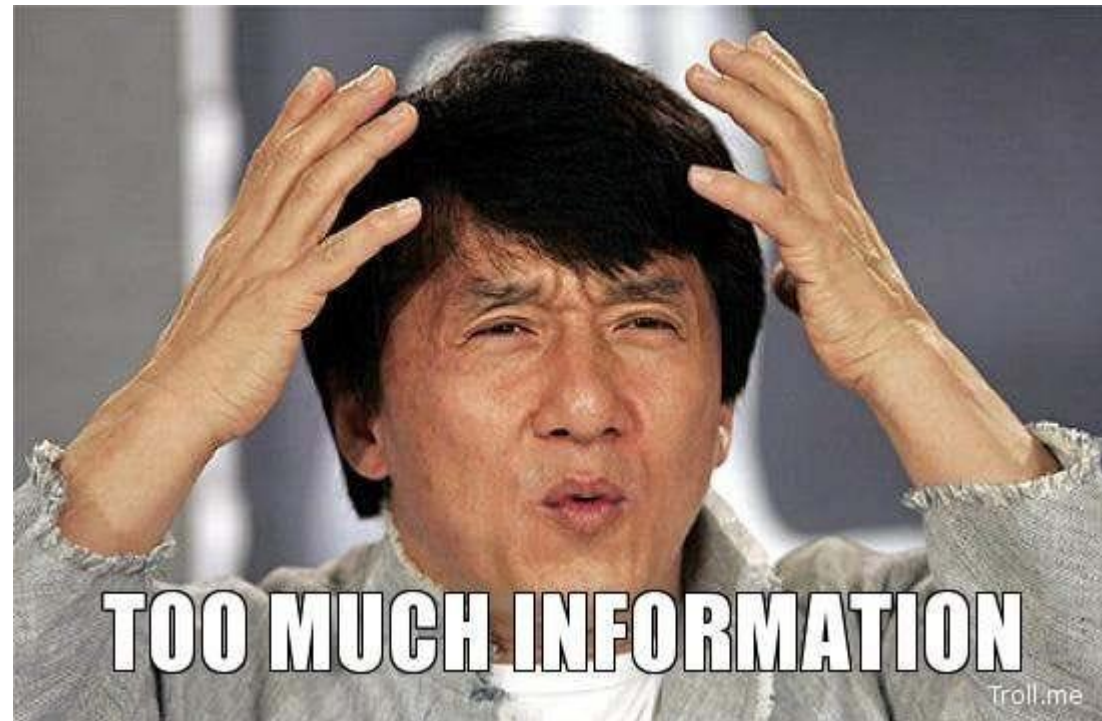
- What is Dimension Reduction
 - Why is it important?
 - When should we use it?
- Dimension Reduction Types
 - Feature Selection
 - Feature Extraction
 - Clustering



Dimension Reduction Is...

Dimension Reduction

The process of reducing the number of input variables (features)...



Dimension Reduction

The process of **reducing the number of input variables (features)...**

while retaining essential information

when you have significant data but no significant other



You are The Dimension Reducer (part 1)

- Choose 5 most important factors/features



Choose 5 to predict mastitis

- Milk Yield (Total)
- Milk Yield in First 2 minutes
- Change From Last Milk Yield (Total)
- Flow rate (0-15 seconds)
- Flow rate (15-30 seconds)
- Change from last flow rate (0-15 seconds)
- Minimum Temperature
- Maximum Temperature
- Sunlight Duration
- Ventilation type used in barn
- Bedding type used in barn
- Lactation Number (how many times has the cow had a baby)

You are The Dimension Reducer (part 2)

- Group similar variables into 1-5 groups



Group variables (Groups predict Mastitis): 1-5 groups

- Milk Yield (Total)
- Milk Yield in First 2 minutes
- Change From Last Milk Yield (Total)
- Flow rate (0-15 seconds)
- Flow rate (15-30 seconds)
- Change from last flow rate (0-15 seconds)
- Minimum Temperature
- Maximum Temperature
- Sunlight Duration
- Ventilation type used in barn
- Bedding type used in barn
- Lactation Number (how many times has the cow had a baby)

Why **select** most important features?

Why **select** most important features?

- Keep original variables
- Remove the variables who do not contribute useful information

Why **select** most important features?

Pros

- Easy to interpret/explain
- Variables keep their original biological meaning
- Reduce redundancy
- Reduces overfitting



Why **select** most important features?

Pros

- Easy to interpret/explain
- Variables keep their original biological meaning
- Reduce redundancy
- Reduces overfitting

Cons

- May not pick the right ones
- Can pick too few or too many
- Can miss hidden biological relationships
- May remove variables that only matter when combined

Why **group** features?

Why **group** features?

- Combines correlated variables
 - Usually groups mimic broader biological concepts
- Uses relationships between variables

Why **group** features?

Pros

- Captures hidden biological processes
- Uses relationships between variables
- Works well with complex systems
- Reduce redundancy
- Reduces overfitting



Why **group** features?

Pros

- Captures hidden biological processes
- Uses relationships between variables
- Works well with complex systems
- Reduce redundancy
- Reduces overfitting

Cons

- Harder to interpret
- Can require stronger statistical justification
- Can pick the wrong number of groups
- Data might not form clear groups

Grouping VS Selecting

Feature Selection

- Keeps individual variables

Grouping

- Combines related variables

Grouping VS Selecting

Feature Selection

- Keeps individual variables
- Easier to interpret

Grouping

- Combines related variables
- Harder to interpret

Grouping VS Selecting

Feature Selection

- Keeps individual variables
- Easier to interpret
- Better for simple prediction

Grouping

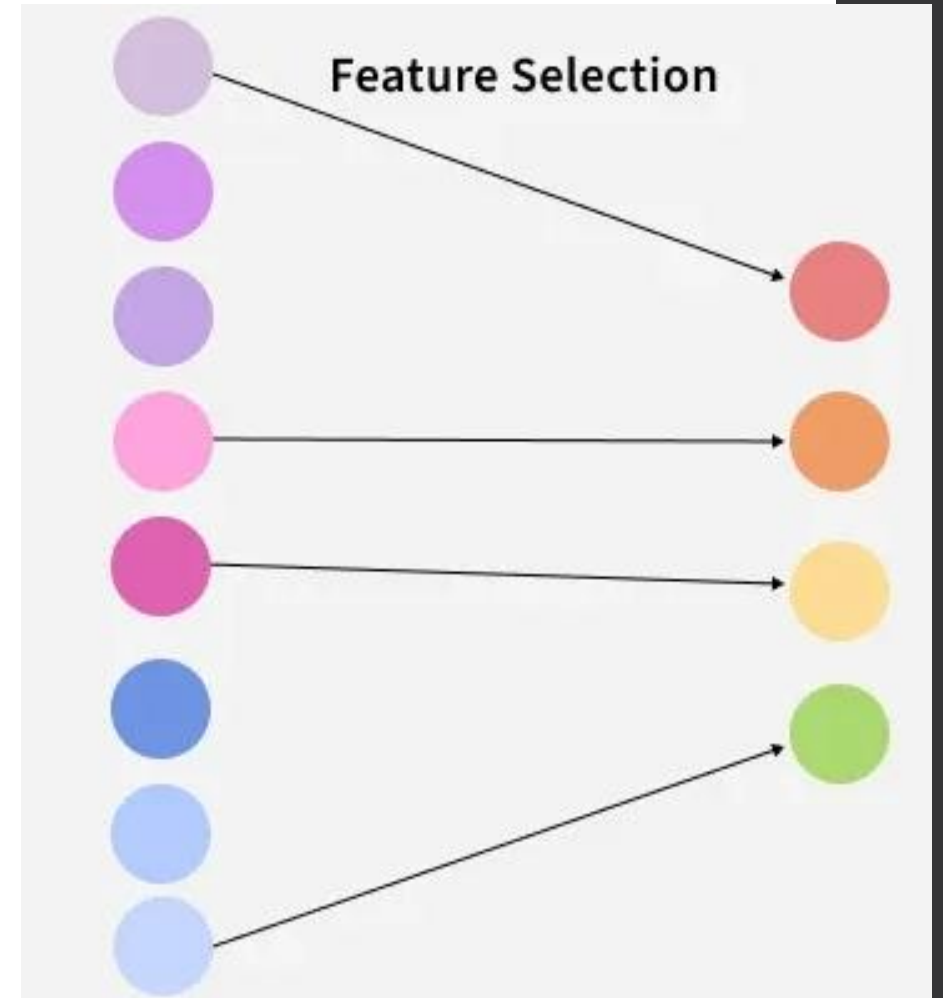
- Combines related variables
- Harder to interpret
- Better for complex prediction

Types of Dimension Reduction

Dimension Reduction Techniques

Selecting

- Feature Selection



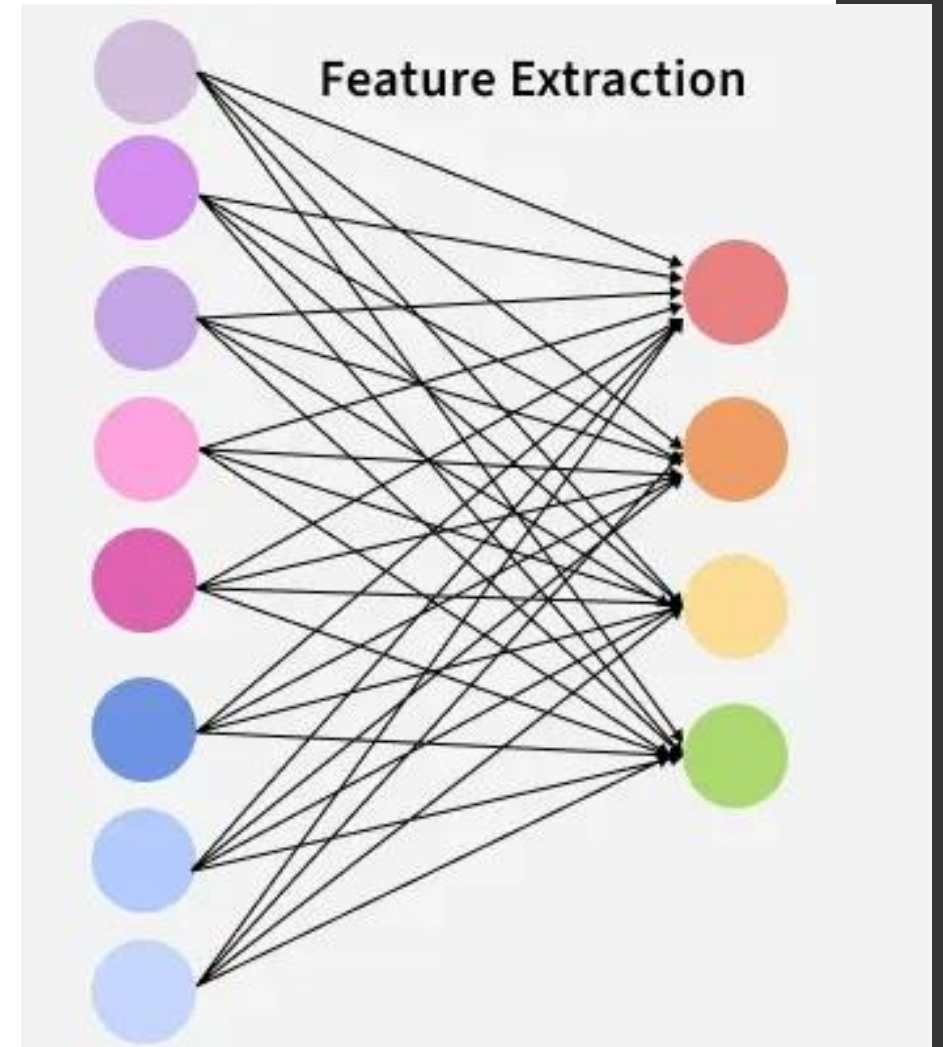
Dimension Reduction Techniques

Selecting

- Feature Selection

Grouping

- Feature Extraction



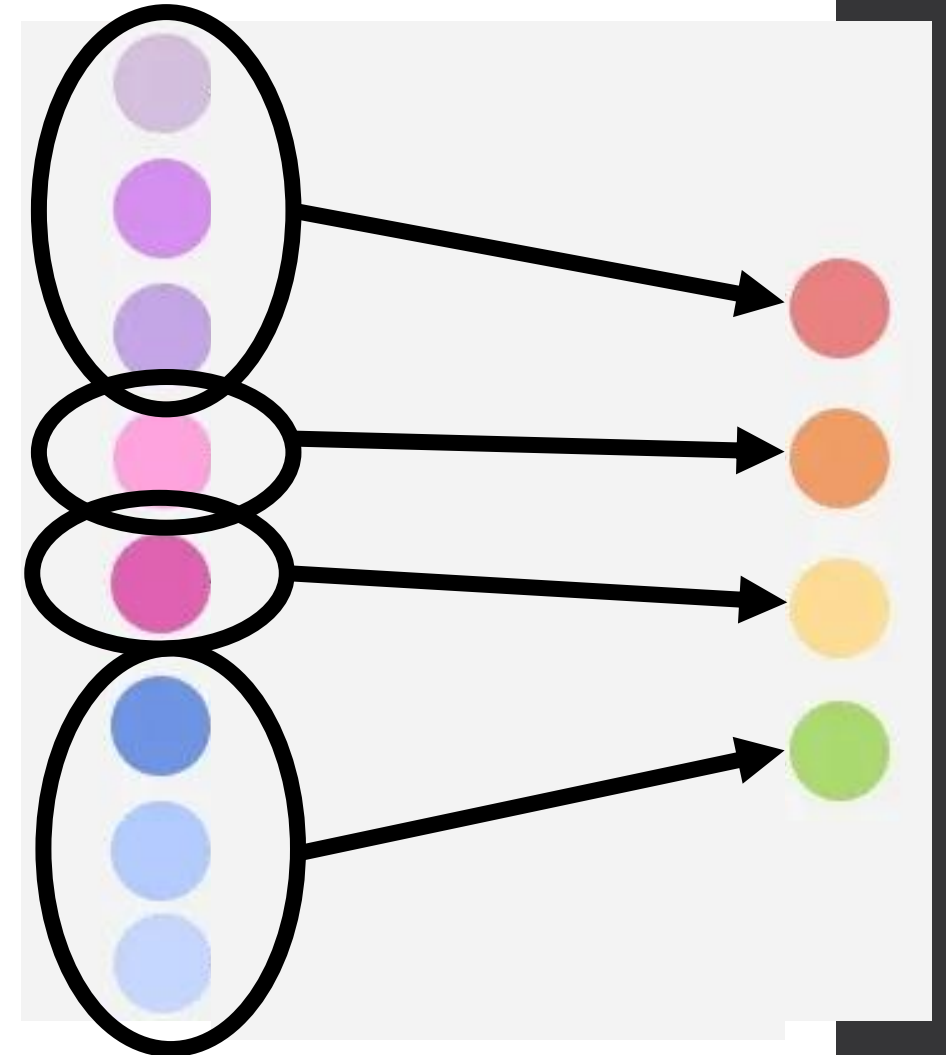
Dimension Reduction Techniques

Selecting

- Feature Selection

Grouping

- Feature Extraction
- Clustering



Dimension Reduction Techniques

Selecting

- Feature Selection

Grouping

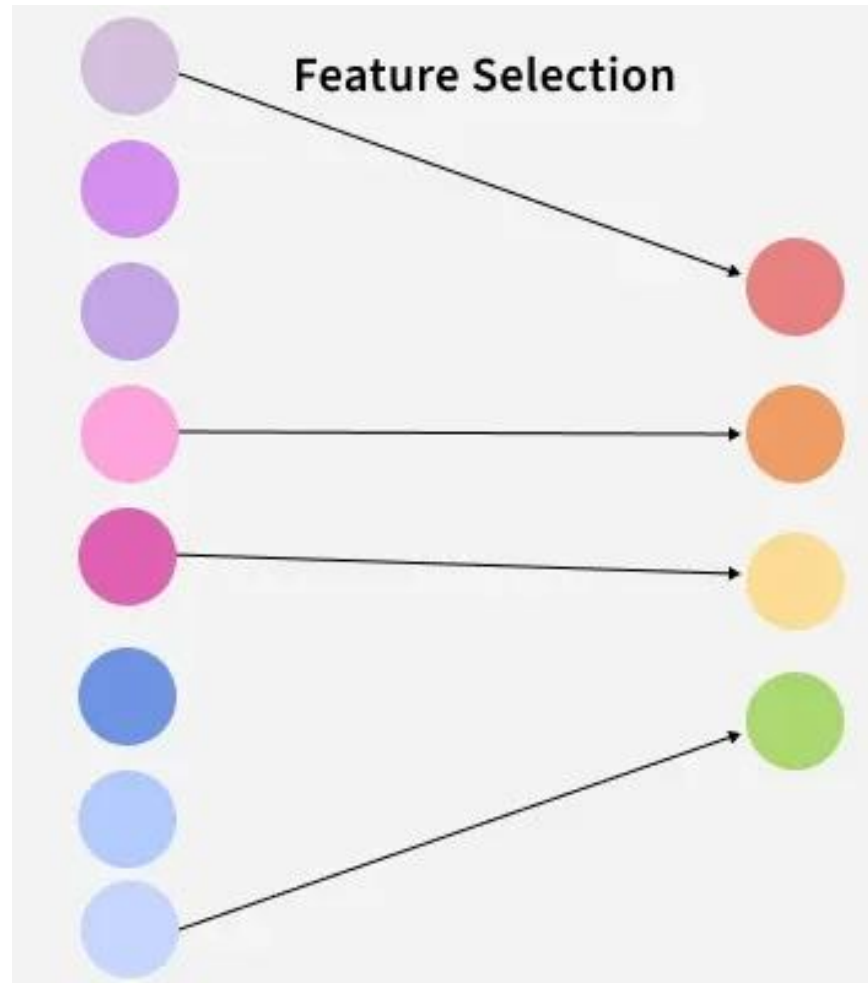
- Feature Extraction
- Clustering
- Combinations are possible!

Selecting

Grouping

Doing
Both





Feature Selection

Feature Selection Options

- **By yourself (Domain/Expert Knowledge)**
- **LASSO**
- **Recursive Feature Elimination**

Feature Selection Options

- **By yourself (Domain/Expert Knowledge)**
- **LASSO** (and other regularization methods)
- **Recursive Feature Elimination**
- Tree-Based (e.g., Random Forest Feature Selection)
- Correlation-Based (Pearson/Spearman Correlation Variance Inflation Factor)
- Univariate Statistical Tests (e.g., t-test, ANOVA, Mann-Whitney U)
- Regressions (e.g., logistic)

Feature Selection: By yourself

- Using an expert's opinion or your own knowledge to **decide what features/variables** to include



Feature Selection: By yourself

Why?

- Easy to explain



Feature Selection: By yourself

Why?

- Easy to explain
- Useful when **prior knowledge** suggests which variables will matter the most



Feature Selection: By yourself

Why?

- Easy to explain
- Useful when **prior knowledge** suggests which variables will matter the most
- Stronger when **limited data** makes machine learning approaches less reliable



Feature Selection: By yourself

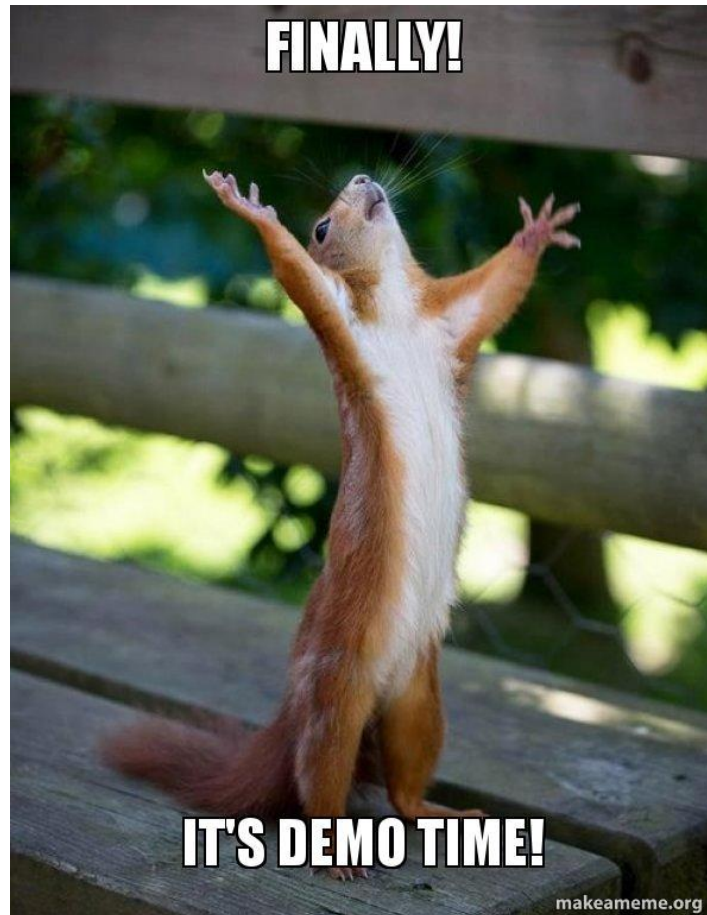
Why?

- Easy to explain
- Useful when **prior knowledge** suggests which variables will matter the most
- Stronger when **limited data** makes machine learning approaches less reliable
- Can prevent models from selecting statistically strong but biologically meaningless variables



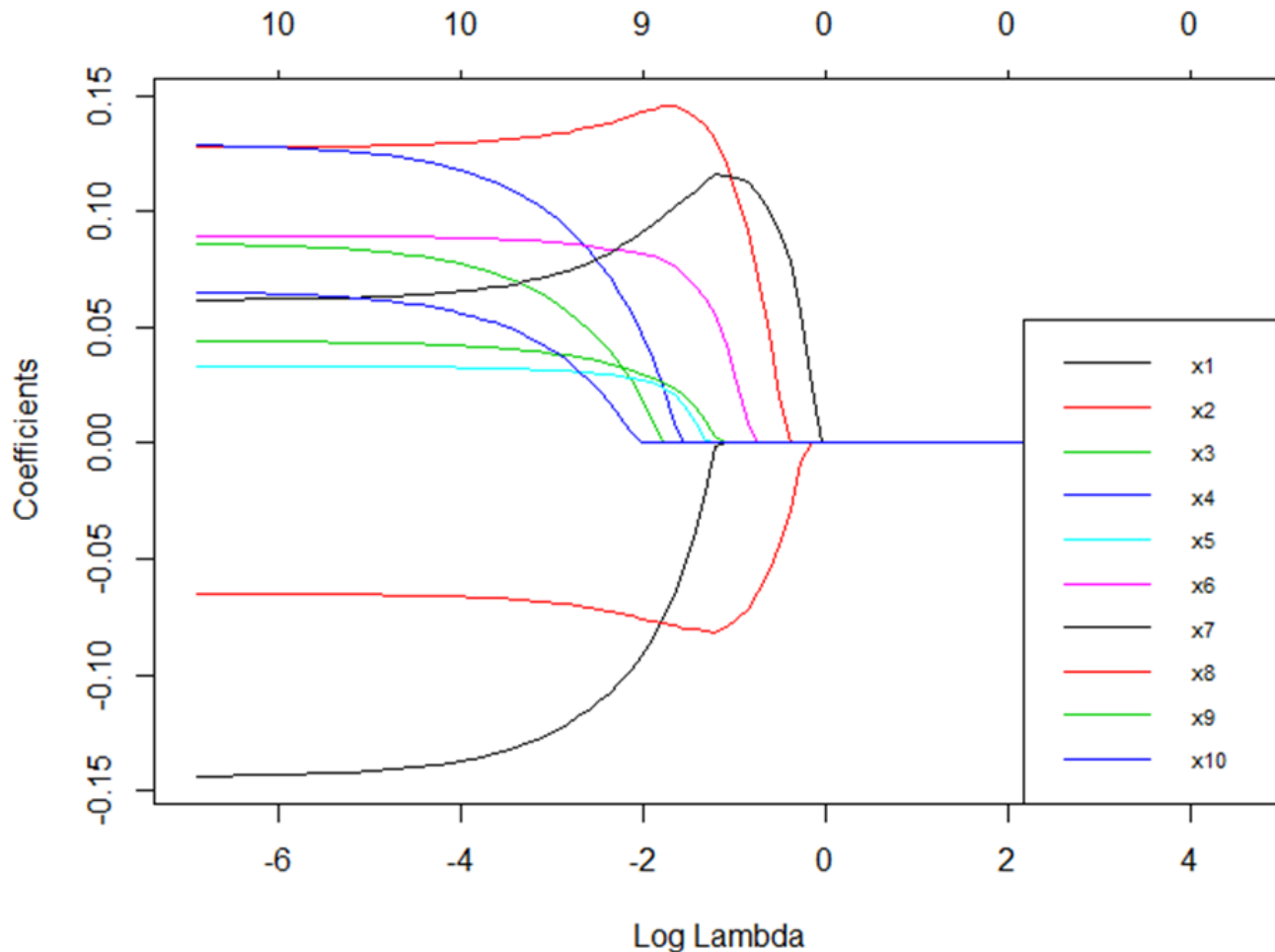
Feature Selection: By yourself

- Demo: How to implement an example in R



Feature Selection: LASSO

The coefficients of the least important factors become 0.



Feature Selection: LASSO

Analogy:

Packing for a trip

- Space is limited
 - The most important items are kept.
 - The less useful ones are left.



Feature Selection: LASSO

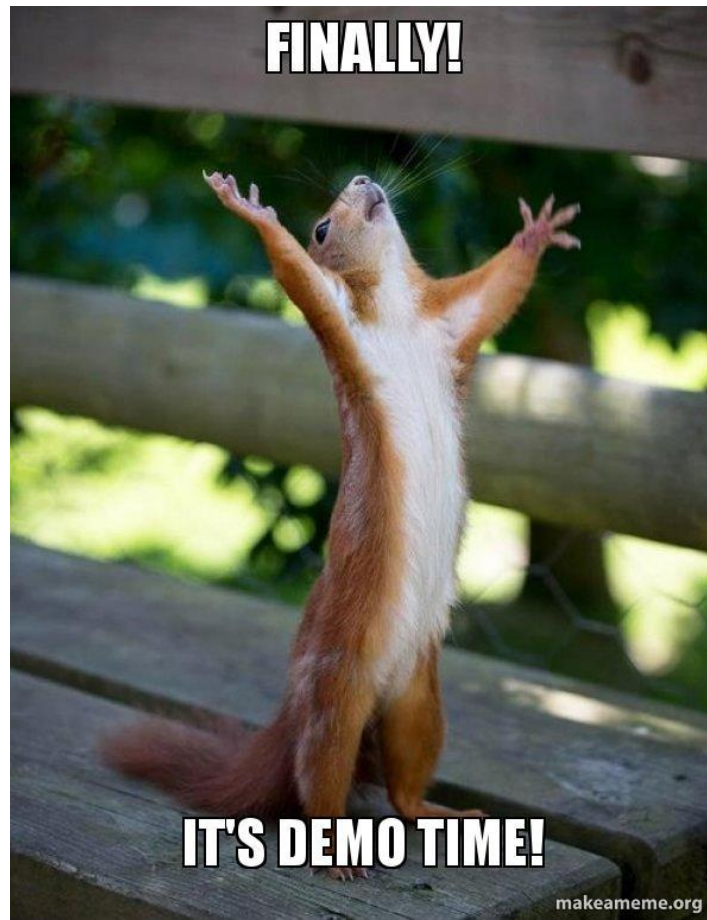
Why?

- Reduces Variance
- Reduces overfitting
- Removes irrelevant features



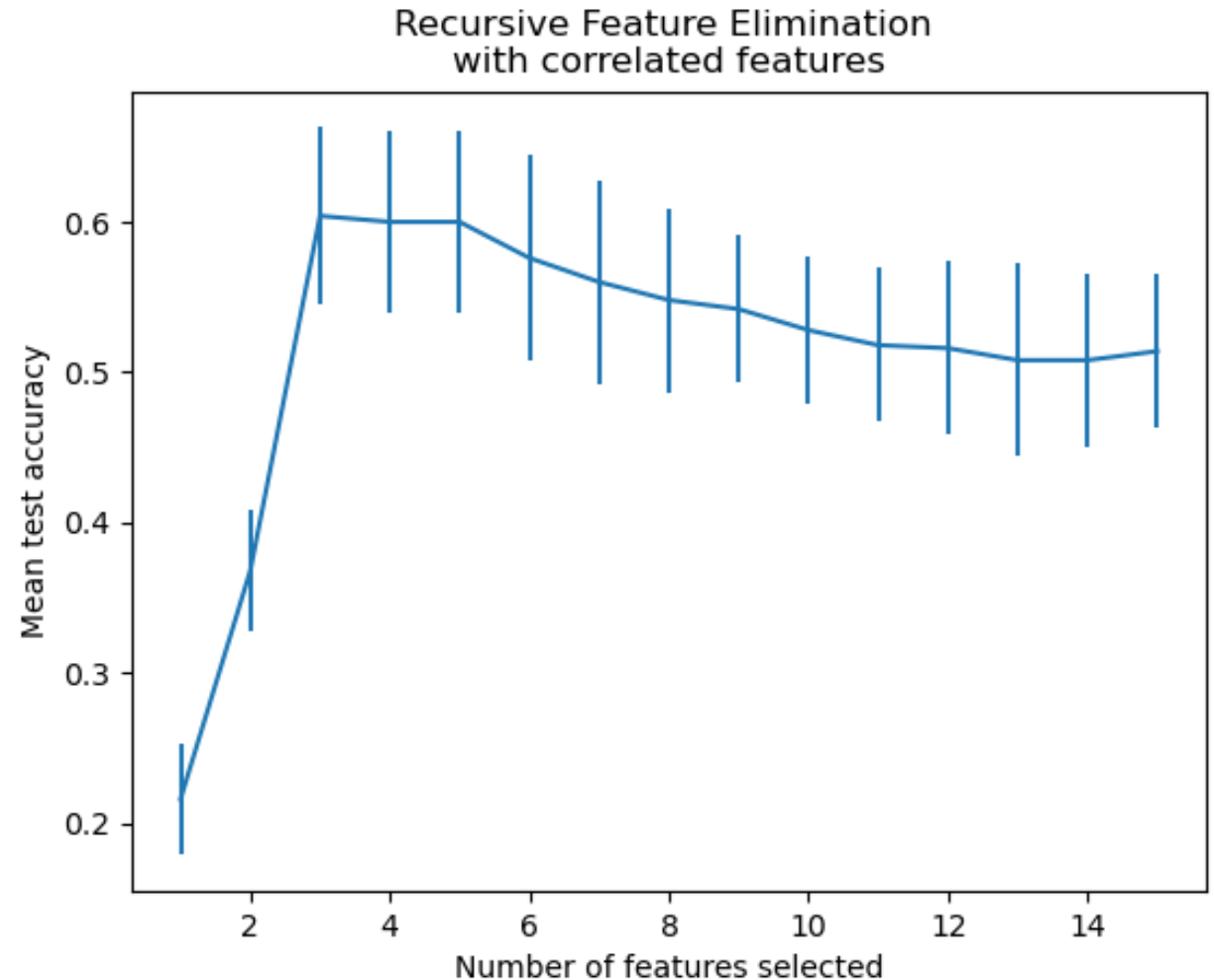
Feature Selection: LASSO

- Demo: How to implement an example in R



Feature Selection: Recursive Feature Elimination

Finds the **best set of variables** by repeatedly **removing the least important** features and checking how the accuracy changes



Feature Selection: Recursive Feature Elimination

Analogy

A tournament for the variables

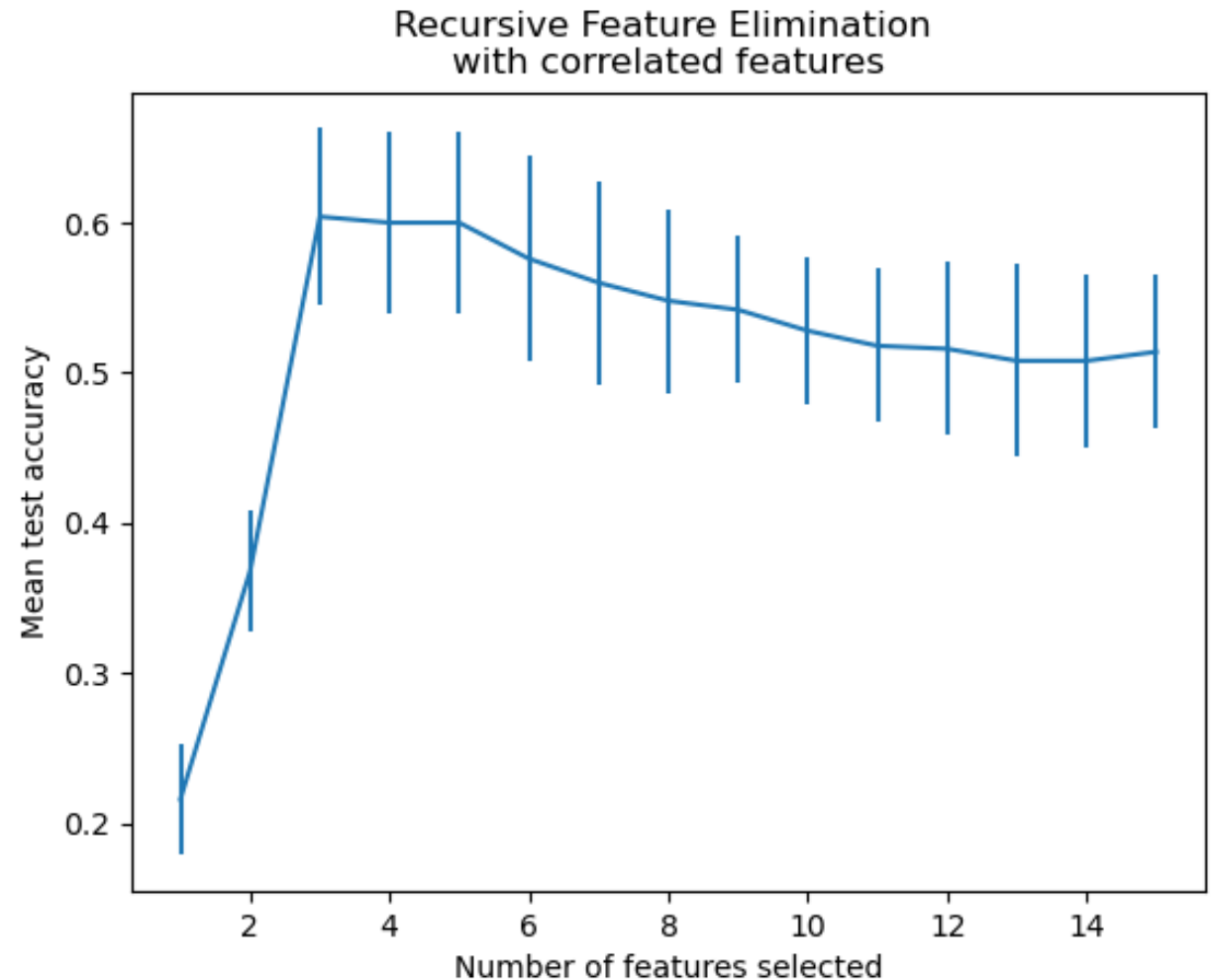
- Weakest features get eliminated one round at a time until the strongest remain



Feature Selection: Recursive Feature Elimination

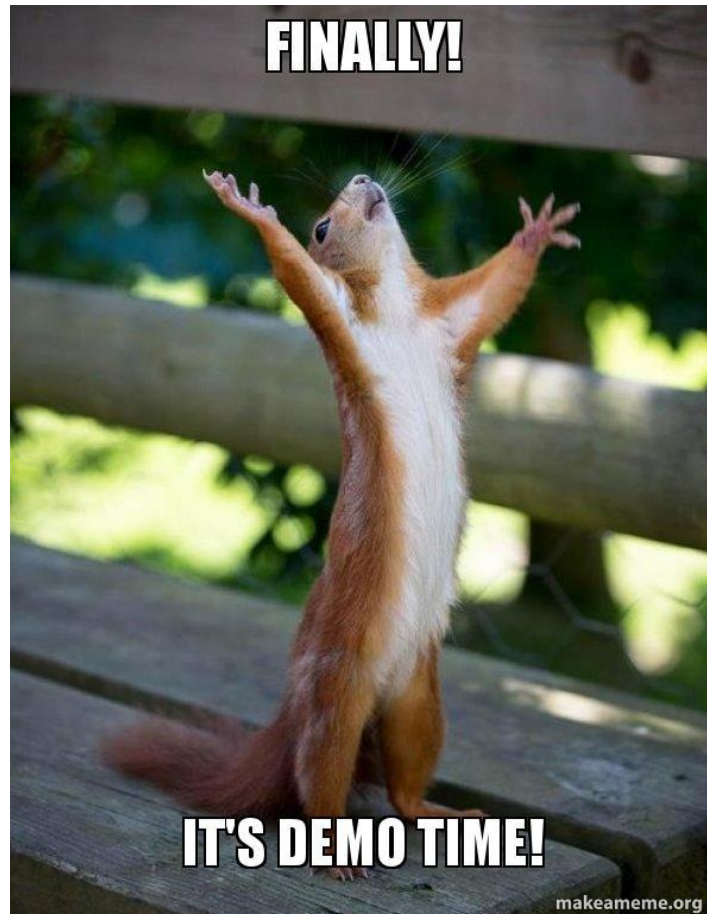
Why?

- Automatically determine the threshold for how many features to include



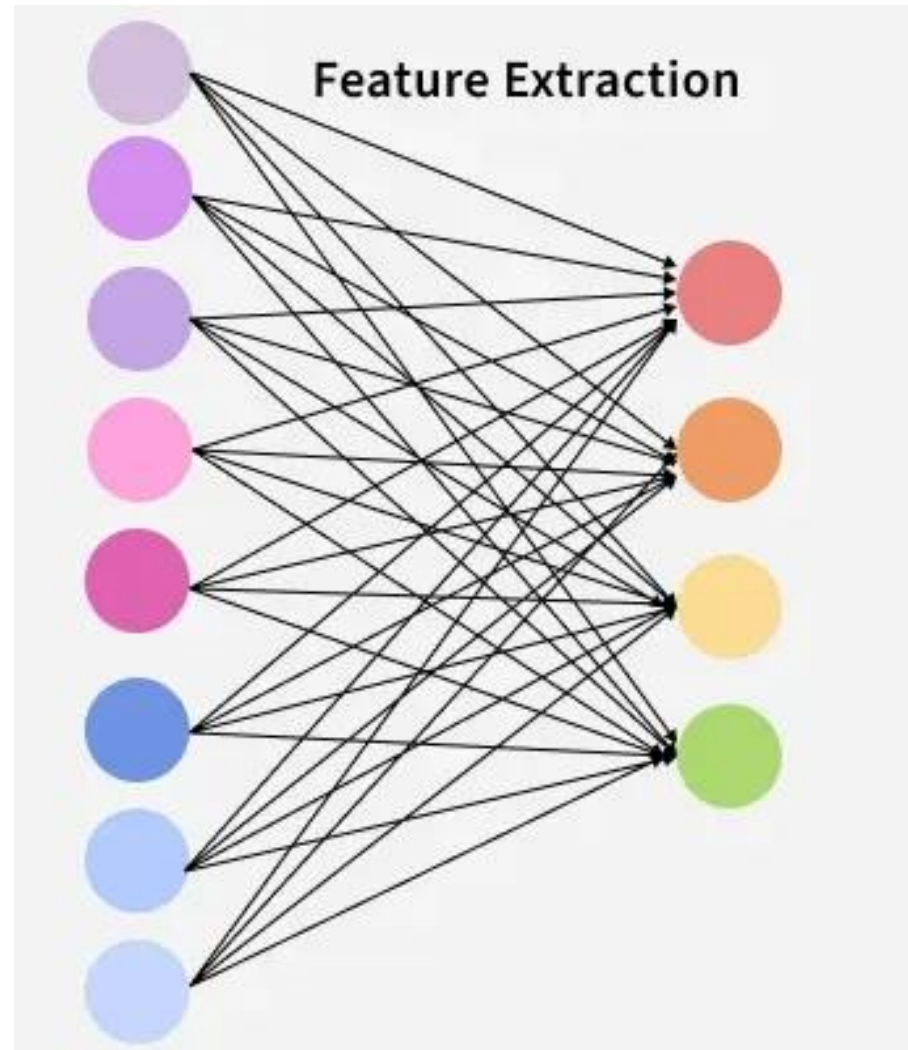
Feature Selection: Recursive Feature Elimination

- Demo: How to implement an example in R



Which Type of Feature Selection

	By Yourself	LASSO	Recursive Feature Elimination
Interpretability	Super Easy	Easy	Somewhat Easy
Clarity of Include vs Exclude	Somewhat clear	Clear	Clear
Statistical rigor	Lower	High	High
Works with small data	Good	Moderate	Moderate
Model Optimization	Lower	Strong	Very strong



Feature Extraction

Feature Extraction Options

- **PCA**
- **Multiple Correspondence Analysis (MCA)**
- **Factor Analysis**

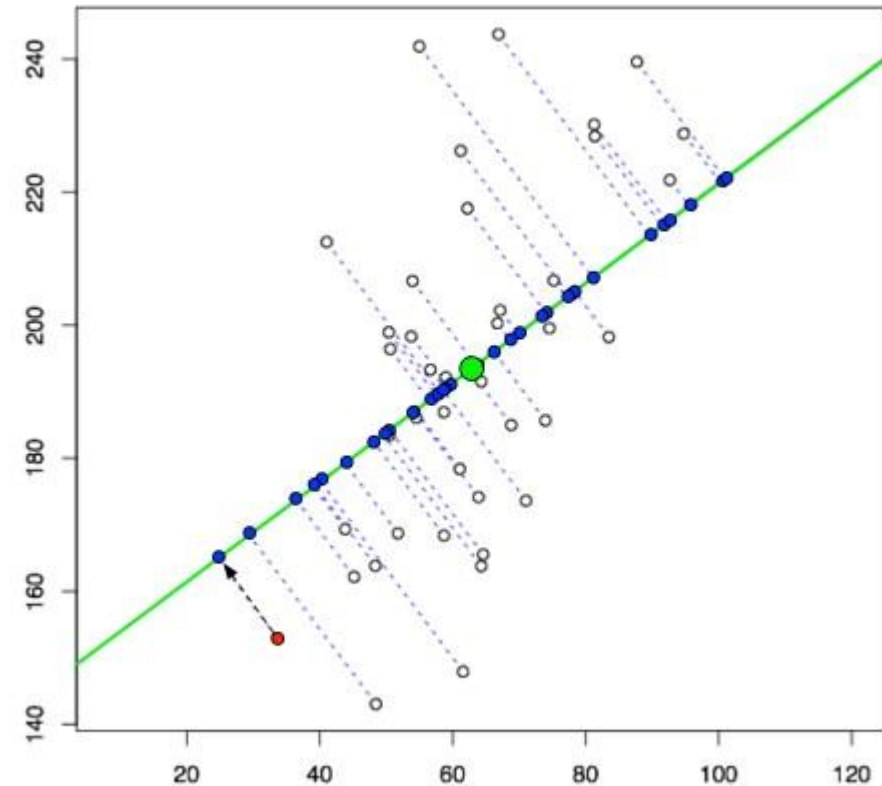
Feature Extraction Options

- **PCA**
- **Multiple Correspondence Analysis (MCA)**
- **Factor Analysis**
- T-SNE (non-linear embedding, function similar to PCA)
- U-MAP
- Partial least squares (extraction + prediction)

Feature Extraction: PCA

PCA (Principal Component Analysis)

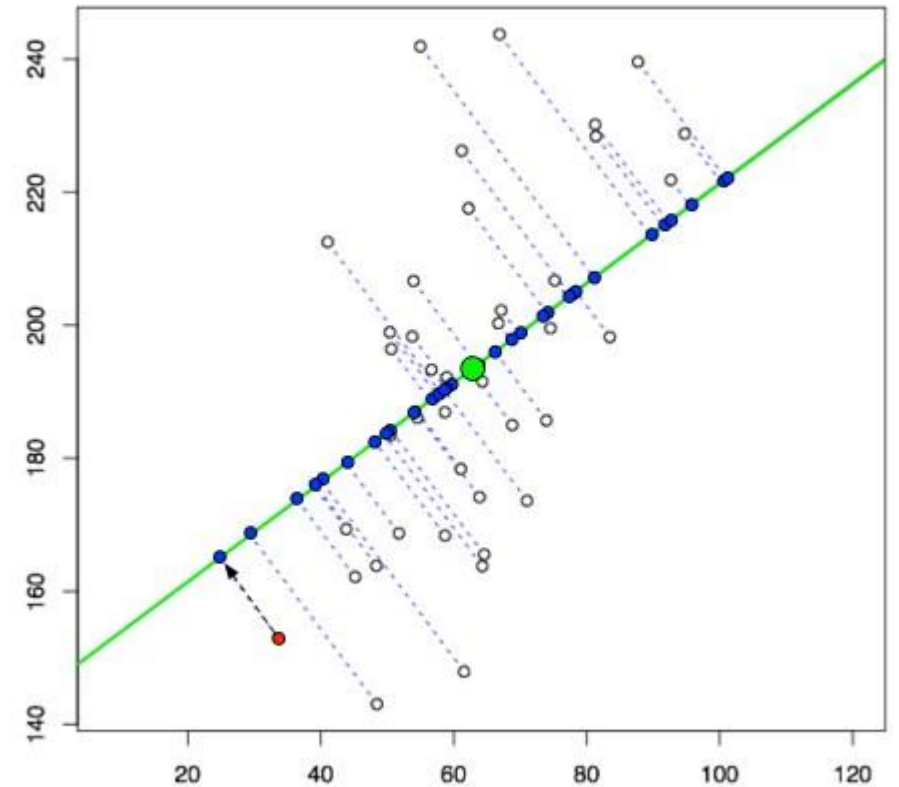
Transforming the data into new features by **maximizing the variance** contained



Feature Extraction: PCA

Further Explained:

PCA takes information from the rows in your data, and...

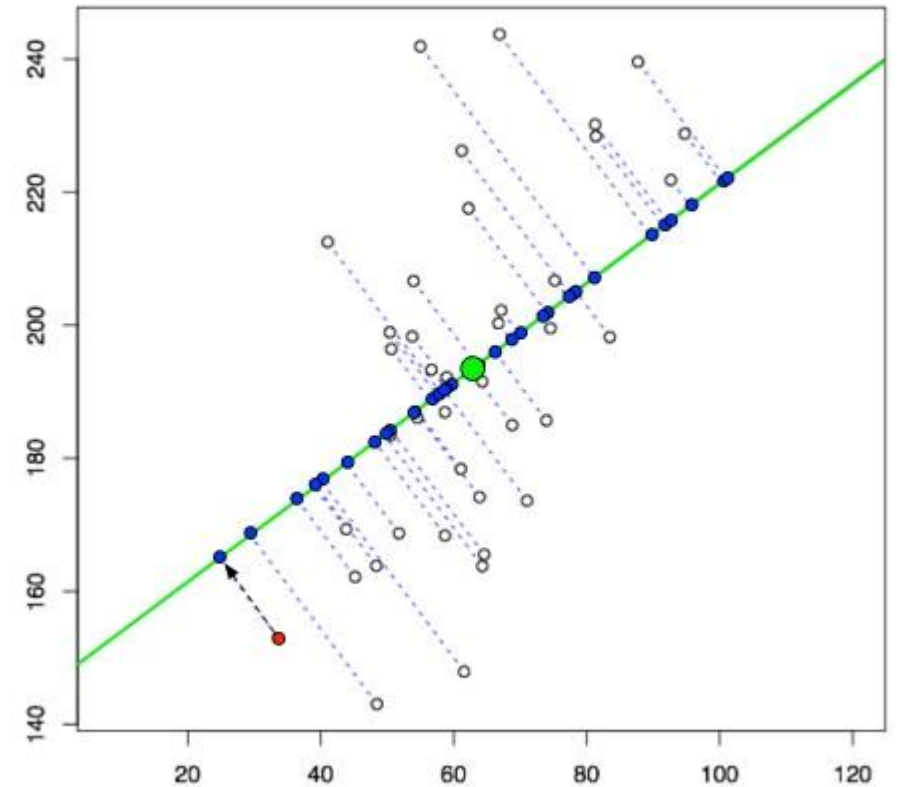


Feature Extraction: PCA

Further Explained:

PCA takes information from the rows in your data, and

1. Looks at the relationships between the rows and columns

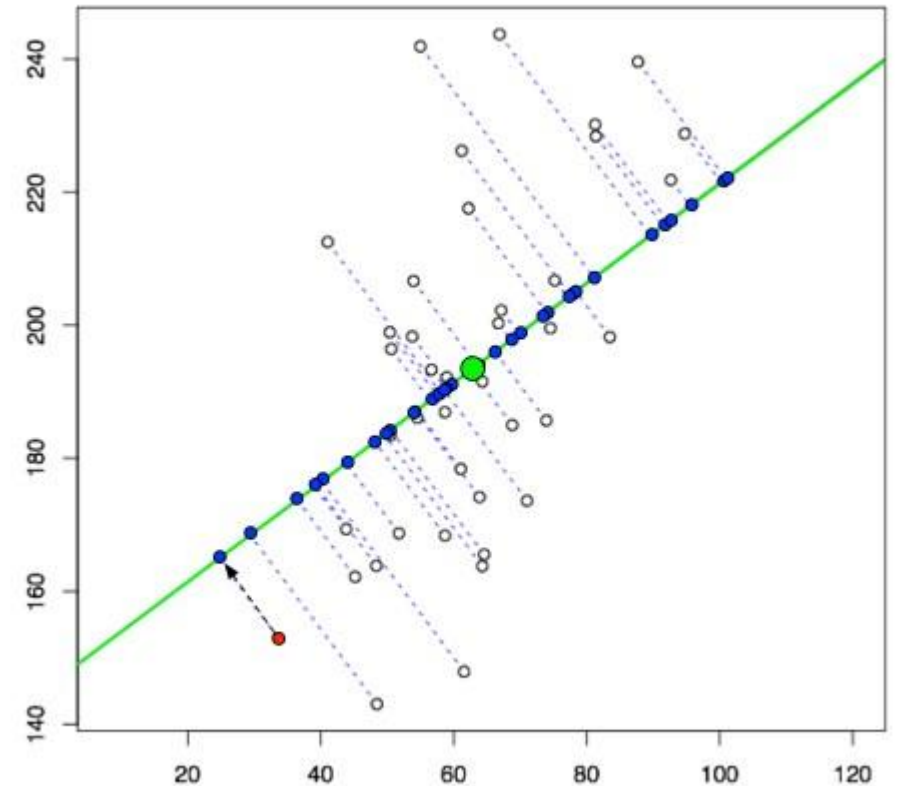


Feature Extraction: PCA

Further Explained:

PCA takes information from the rows in your data, and

1. Looks at the relationships between the rows and columns
2. Decides how to transform your data into a smaller number of new variables called principal components



Feature Extraction: PCA

Analogy

Start with

- Bananas, Strawberries, Grapes, and Lemons

Blend them together to create something new

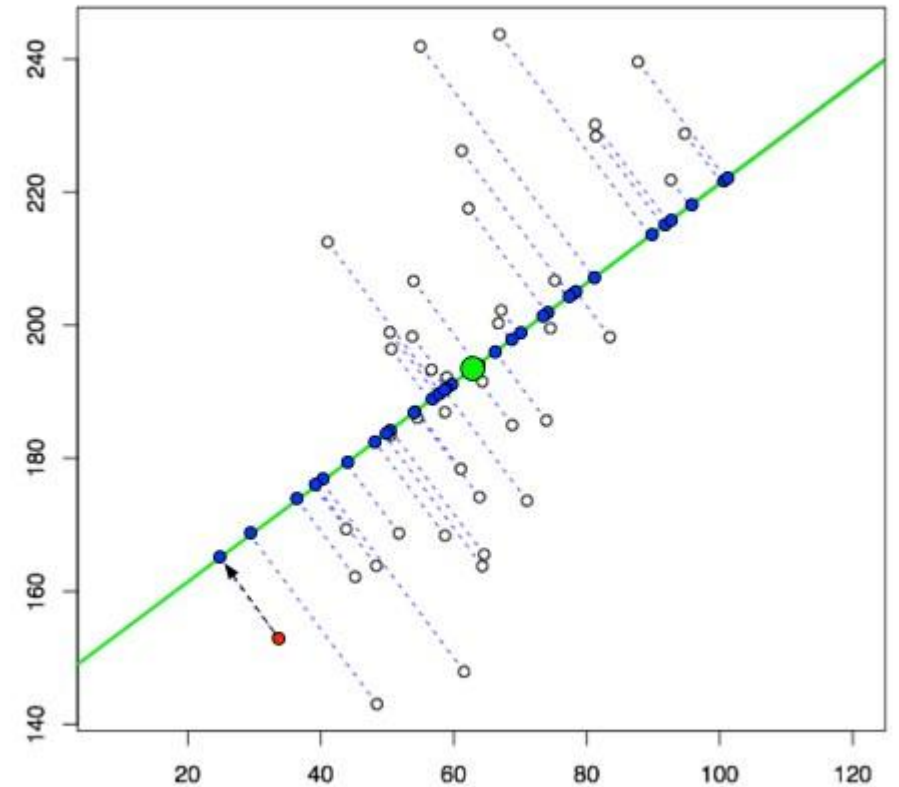
- Keeps major flavors (some more powerful than others)
- **Banana has the strongest “variance” and is the easiest to taste; grapes disappear almost entirely**



Feature Extraction: PCA

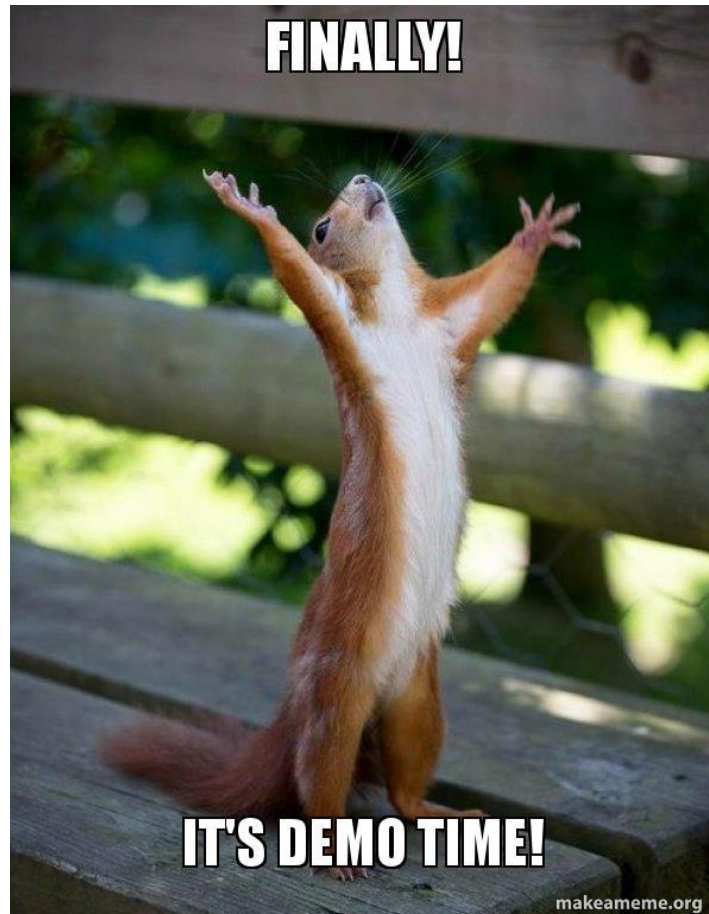
Why?

- Data is complex
- Hidden or unknown patterns exist
- Overlapping variables/multicollinearity is present, but individual variables are difficult to pick



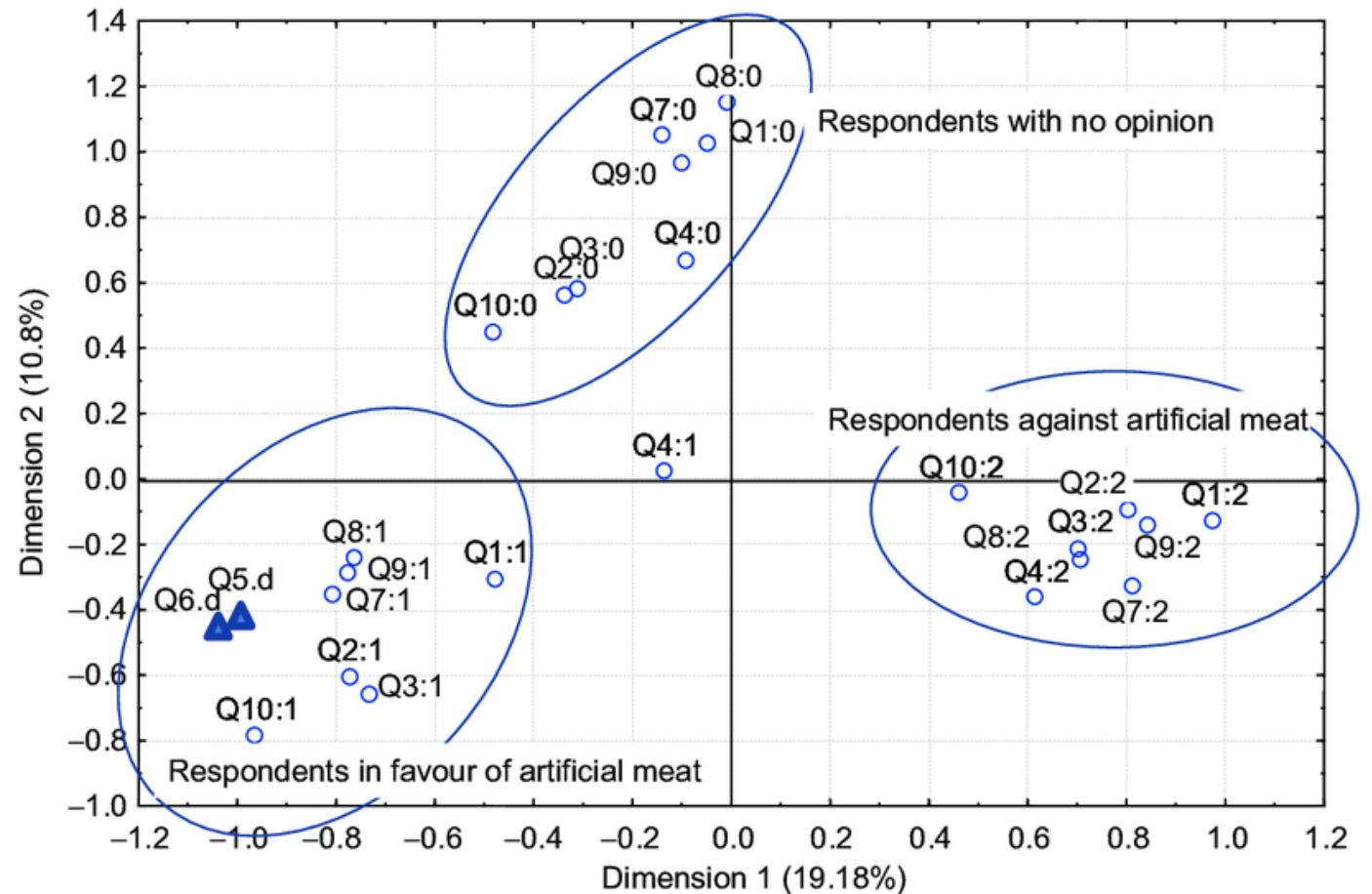
Feature Extraction: PCA

- Demo: How to implement an example in R



Feature Extraction: Multiple Correspondence Analysis

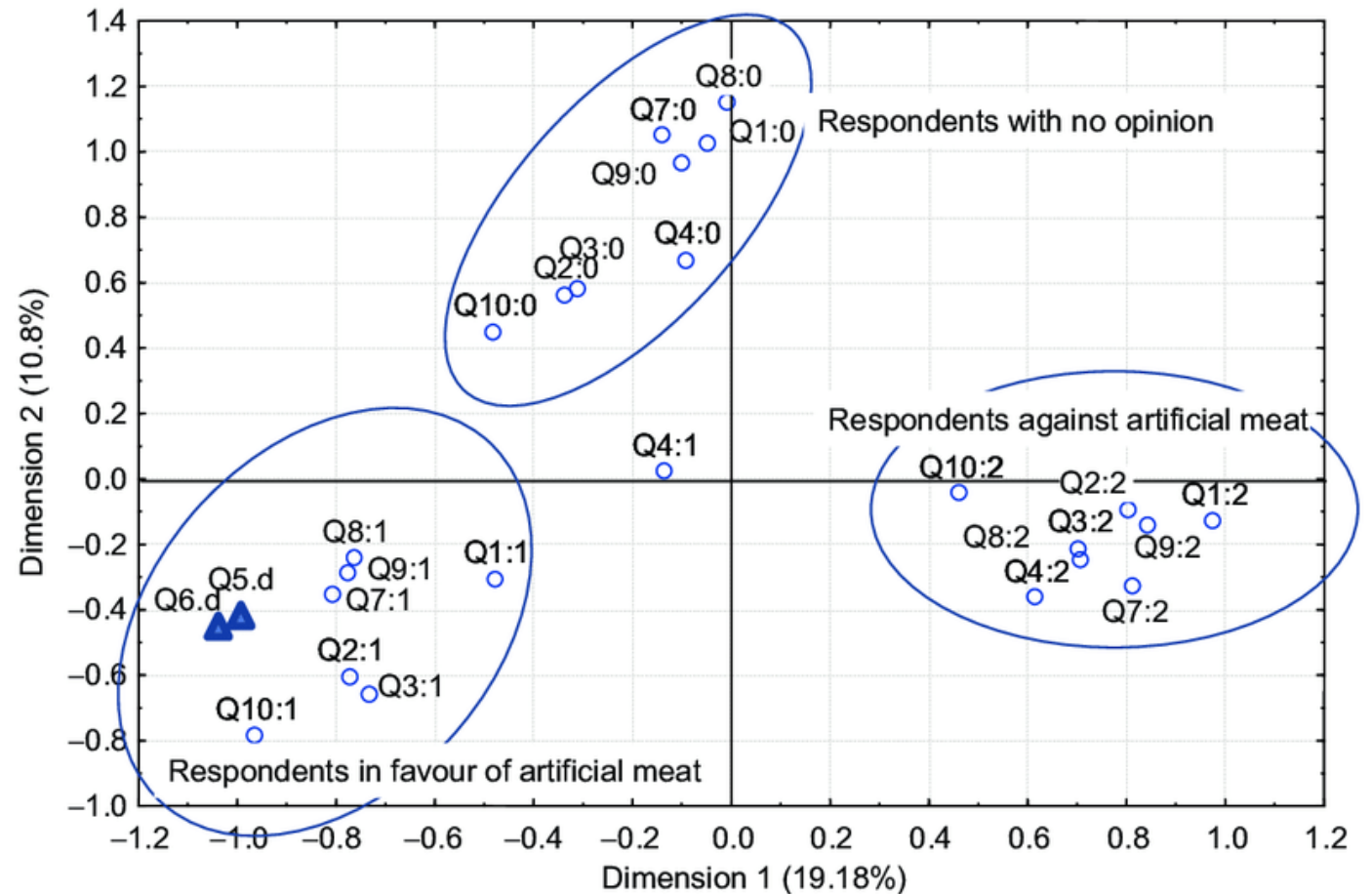
- The PCA for categorical data
- Used to analyze the relationships between several categorical variables



Feature Extraction: Multiple Correspondence Analysis

Why?

- Analyzing complex categorical data
- Hidden or unknown patterns exist
- Overlapping variables are present, but individual variables are difficult to pick



Feature Extraction: Multiple Correspondence Analysis

No demo in R

<https://www.sthda.com/english/articles/31-principal-component-methods-in-r-practical-guide/114-mca-multiple-correspondence-analysis-in-r-essentials/>

Feature Extraction: Factor Analysis

Statistical technique to **use the correlation and variance** between variables to reduce variables **into underlying patterns/factors**.

- Assumes normality if the data is numeric



Feature Extraction: Factor Analysis

Why?

- Multiple observed variables reflect an underlying latent factor
- Reveals hidden relationships among correlated variables
- Useful when direct measurement is impossible, but indirect indicators are available



Feature Extraction: Factor Analysis

Example

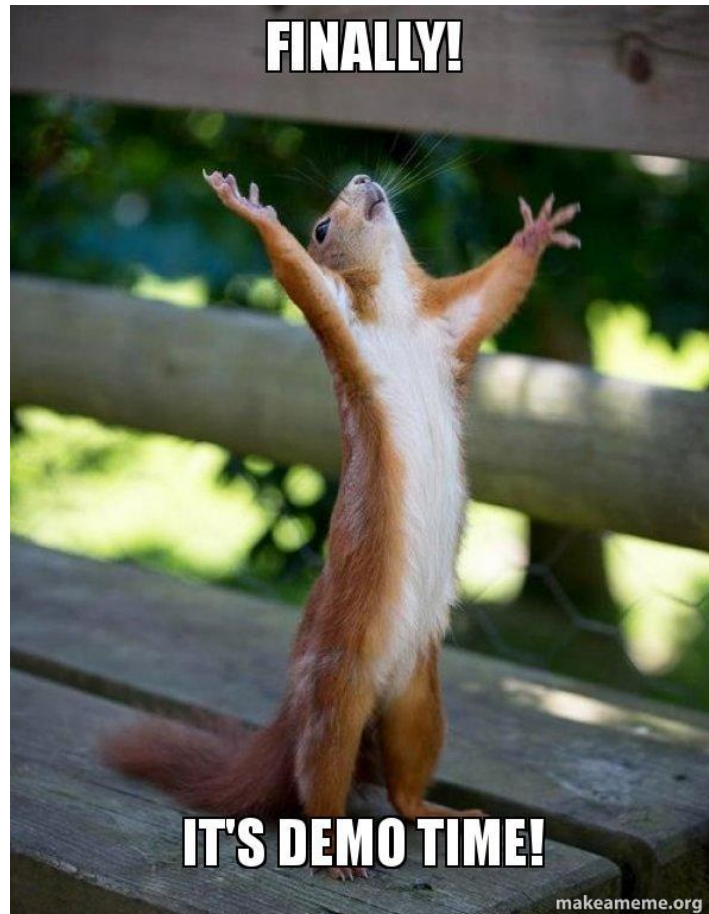
- Dog is having...
 - High waste levels + water intake
 - Loss of appetite
 - Vomiting and Diarrhea
 - Bad Breath
 - Pale gums

Underlying factors of Kidney disease



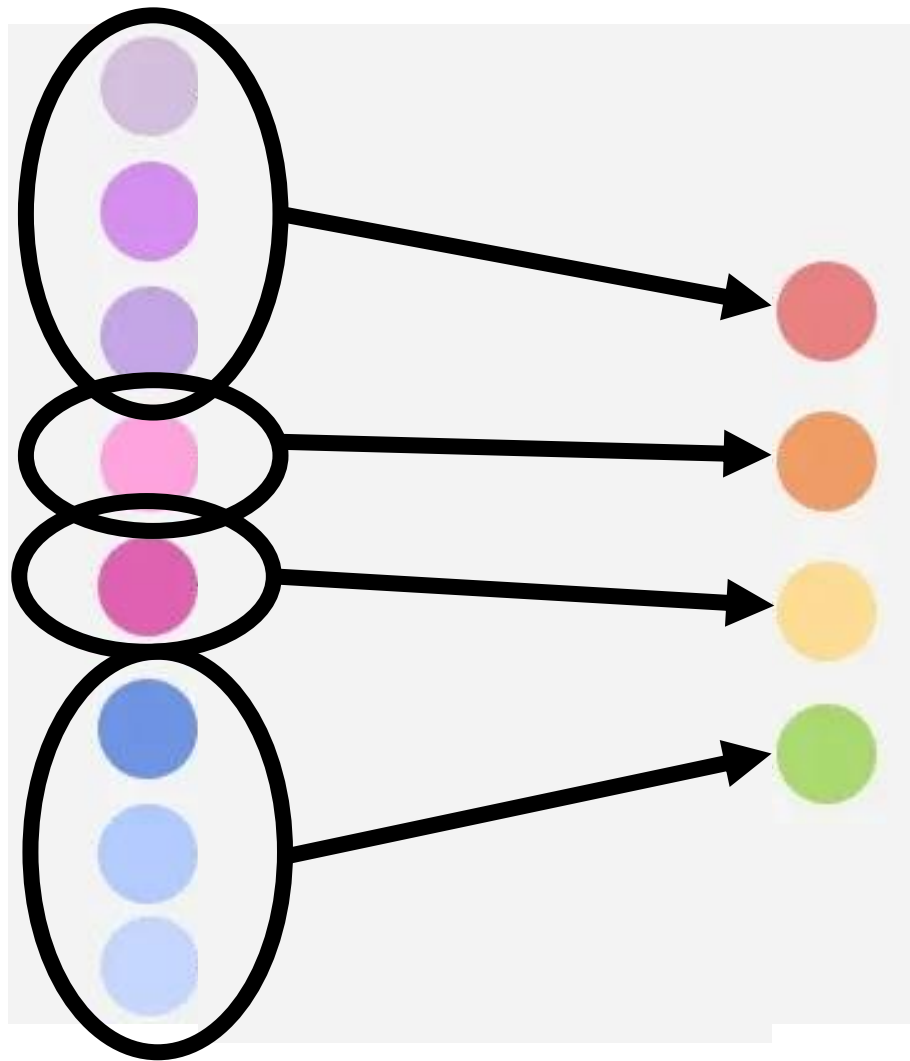
Feature Extraction: Factor Analysis

- Demo: How to implement an example in R



Which Type of Feature Extraction

	PCA	Multiple Correspondence Analysis	Factor analysis
Data Type	Numeric	Categorical	Numeric, binary, ordered categories
Explaining...	Variance + Finding patterns	Variance + Finding Patterns	Underlying biology + Explaining Patterns
Works with small data	Not really	Not really	Moderately



Clustering

Clustering Options

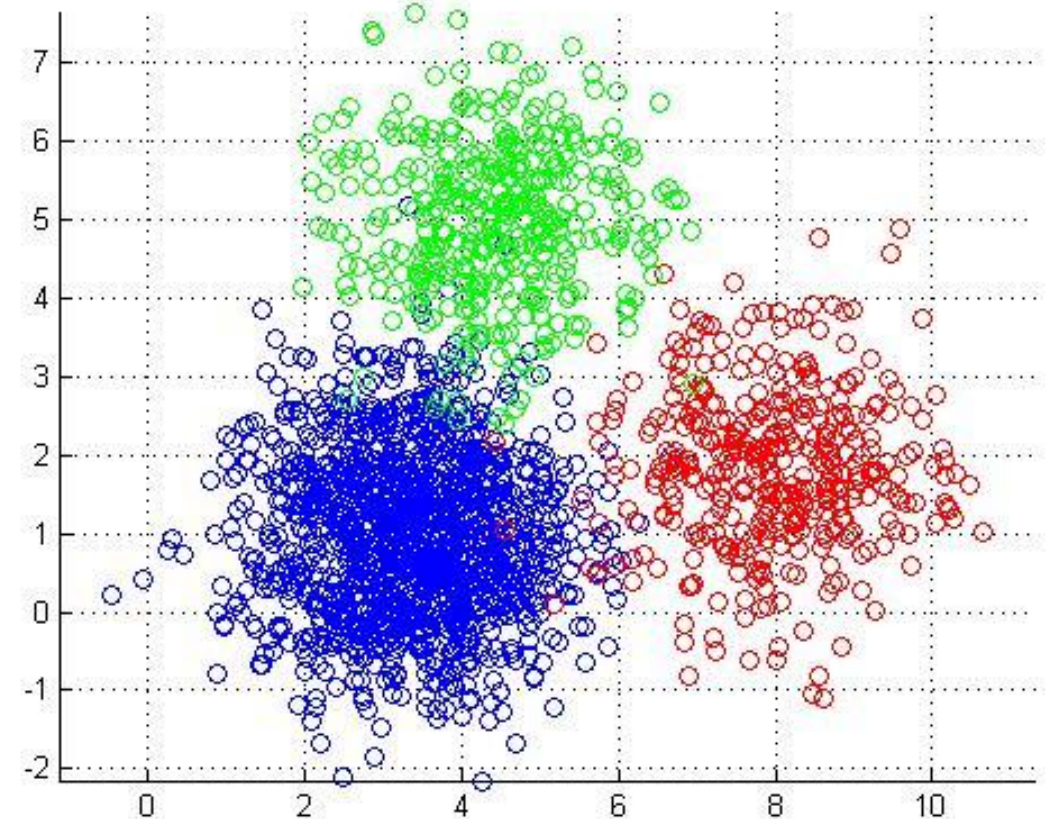
- **K-Means Clustering**
- **Latent Class Analysis**

Clustering Options

- **K-Means Clustering**
- **Latent Class Analysis**
- Tree-Based Clustering (e.g., Hierarchical Clustering)
- Density-Based Clustering
- Distribution-Based Clustering

Clustering: K-means

- Partition data into “k” distinct clusters by **minimizing the distance between data points and the cluster’s center**
- Group rows/observations together based on how alike they are



Clustering: K-means

Analogy

Imagine sorting a fruit basket.

You have green apples, red apples, oranges, and bananas.

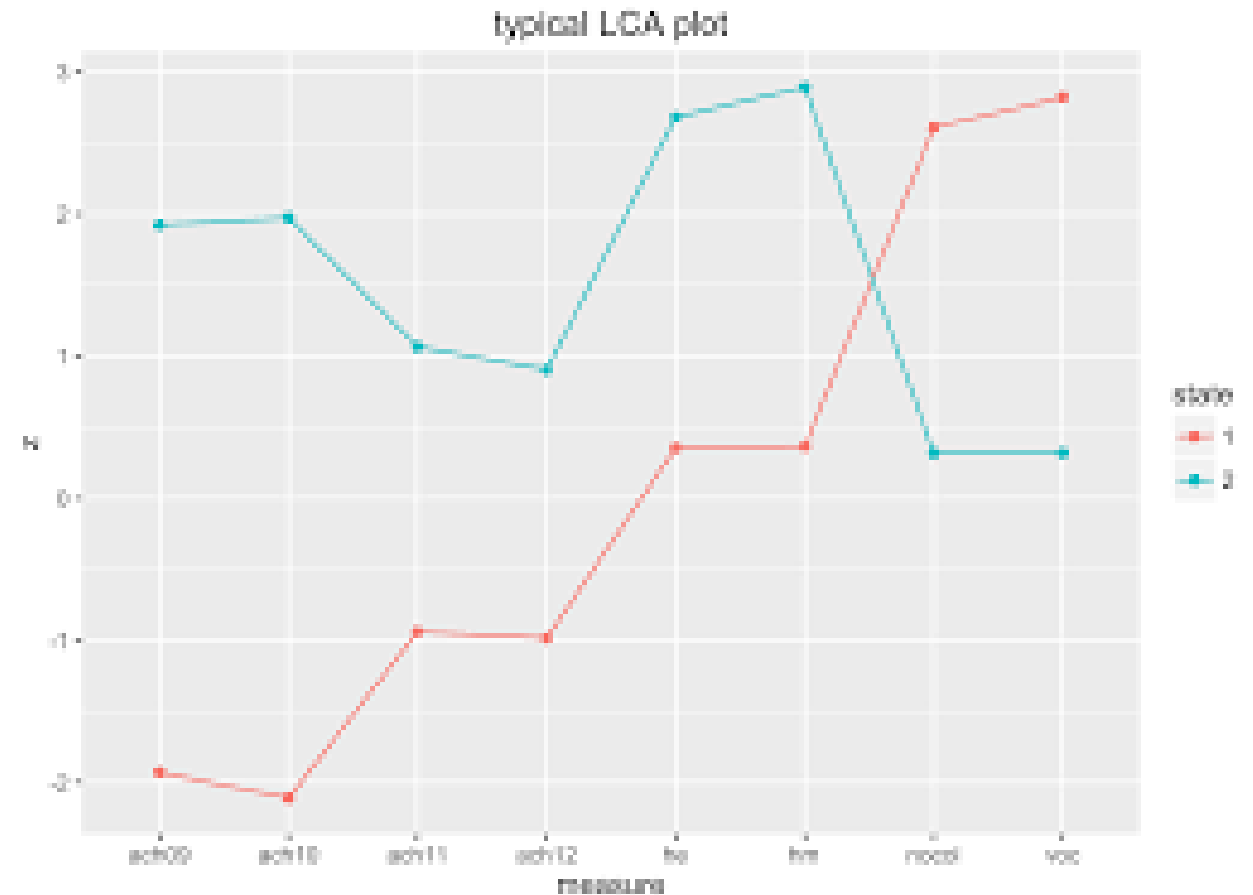
You make 3 piles based on what looks similar

1. Apples (red + green)
2. Oranges
3. Bananas



Clustering: Latent Class Analysis

- Find unobserved (latent) groups based on patterns across multiple variables



Clustering: Latent Class Analysis

Analogy

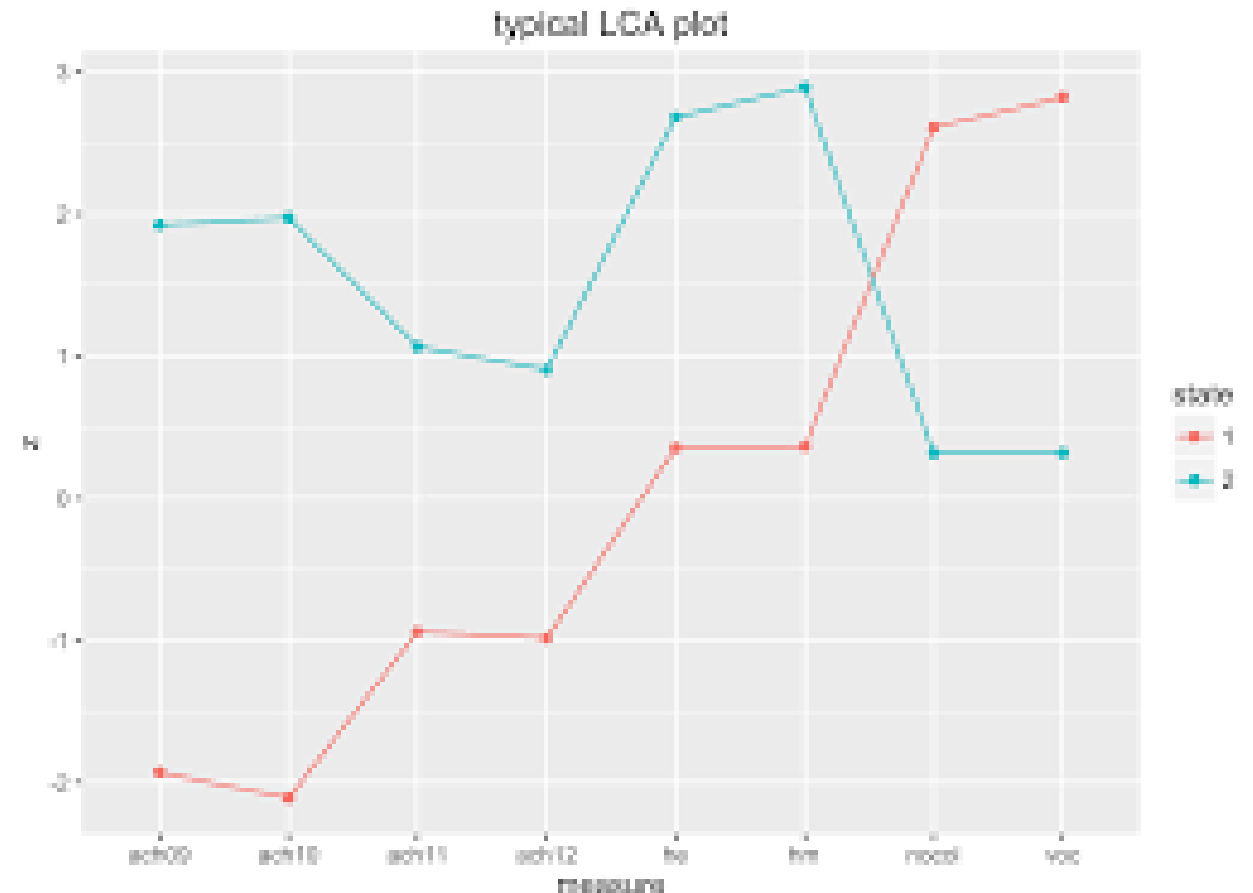
- You are trying to guess who is a (1) Pre-med, (2) Engineering, or (3) Business major
- By looking at patterns like:
 - Backpack type
 - Outfit
 - Coffee order
 - Sleep Schedule
 - Glasses
 - Amount of Hair gel
 - Desperation in eyes



Clustering: Latent Class Analysis

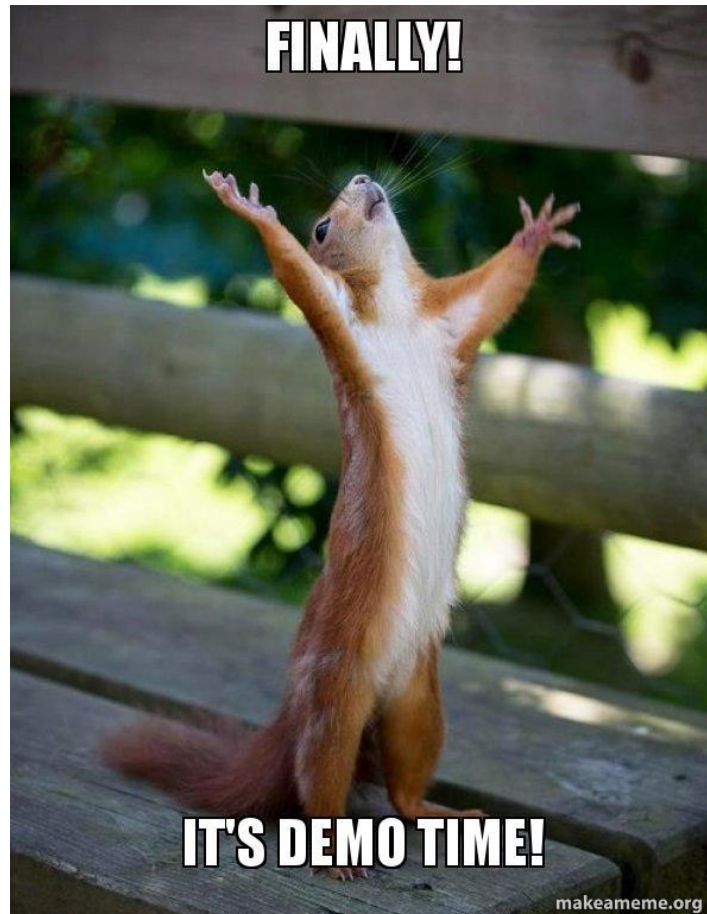
Why?

- Discover hidden subgroups
- Works well with clear (even if unknown) stratification
- Usually more connected to biological data



Clustering: Latent Class Analysis

- Demo: How to implement an example in R



Which Type of Clustering

	K-Means	Latent Class Analysis
Data Type	Mostly numeric	Categorical or mixed
Grouping type	Distance based (fast and practical)	Hypothesis-driven biological grouping
Works with small data	Not really	No

Review Time

Dimension Reduction Is...

Making The Right Decisions

Wrong VS Right

Rule of Thumb Table

Goal	Use
Be extremely interpretable	

Rule of Thumb Table

Goal	Use
Be extremely interpretable	Feature Selection

Rule of Thumb Table

Goal	Use
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Small dataset with strong domain knowledge	

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Rule of Thumb Table

Goal	Use
Be extremely interpretable	Feature Selection
Small dataset with strong domain knowledge	Feature Selection
If there are hidden patterns	

Rule of Thumb Table

Goal	Use
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Small dataset with strong domain knowledge	Feature Selection
If there are hidden patterns	Feature Extraction

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Reduce highly correlated or overlapping variables	

Rule of Thumb Table

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Rule of Thumb Table

Goal	Use
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If there are hidden patterns	Feature Extraction
Reduce highly correlated or overlapping variables	Feature Extraction
Only have indirect observations	

Rule of Thumb Table

Goal	Use
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Rule of Thumb Table

Goal	Use
Be extremely interpretable	Feature Selection
Small dataset with strong domain knowledge	Feature Selection
If there are hidden patterns	Feature Extraction
Reduce highly correlated or overlapping variables	Feature Extraction
Only have indirect observations	Feature Extraction
Identify hidden groupings	

Rule of Thumb Table

Goal	Use
Be extremely interpretable	Feature Selection
Small dataset with strong domain knowledge	Feature Selection
If there are hidden patterns	Feature Extraction
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Only have indirect observations	Feature Extraction
Identify hidden groupings	Clustering

Rule of Thumb Table

Goal	Use
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Only have indirect observations	Feature Extraction
Identify hidden groupings	Clustering
Every Feature matters but unknown how much	

Rule of Thumb Table

Goal	Use
Be extremely interpretable	Feature Selection
Small dataset with strong domain knowledge	Feature Selection
If there are hidden patterns	Feature Extraction
Reduce highly correlated or overlapping variables	Feature Extraction
Only have indirect observations	Feature Extraction
Identify hidden groupings	Clustering
Every Feature matters but unknown how much	Feature Extraction or Clustering

You Pick!

What method(s) to use for Mastitis Prediction?

- Milk Yield (Total)
- Milk Yield in First 2 minutes
- Change From Last Milk Yield (Total)
- Flow rate (0-15 seconds)
- Flow rate (15-30 seconds)
- Change from last flow rate (0-15 seconds)
- Minimum Temperature
- Maximum Temperature
- Sunlight Duration
- Ventilation type used in barn
- Bedding type used in barn
- Lactation Number (how many times has the cow had a baby)

What method(s) to use for Wildlife Disease Surveillance for Storks?

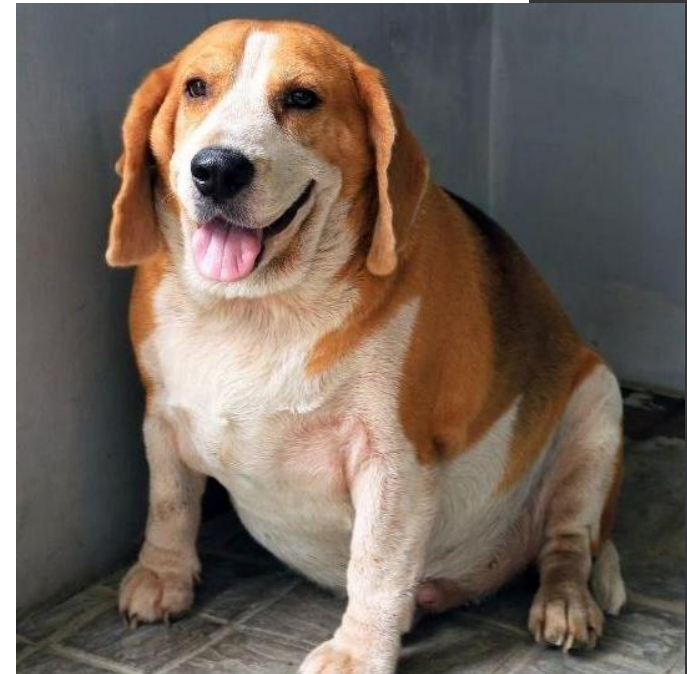
- Temperature
- Rainfall
- Land cover
- Migration routes (done through daily location of 100 randomly sampled storks)
- Monthly estimates of Population density
- Water proximity
- Vector abundance
- Elevation
- Soil moisture

What method(s) for an Antimicrobial Resistance Study

- Average usage of 10 different antimicrobials per week per county in NY (1 feature for each antimicrobial)
- Individual farm management practices (manure management, bedding/housing type, etc.)
- Individual farm's biosecurity measures
- Individual farm's random sample for microbes
- Individual farm's treatment history (veterinary and on-farm records)

What method(s) for a Companion Animal Obesity Study

- Diet
- Exercise time per day
- Breed
- Average time the owner spends away from the house
- Feeding frequency
- Household structure
- Medication use
- Neuter status
- Medical history



What method(s) to use for Clinical Pathology Panels

- Bloodwork
 - ALT
 - AST
 - BUN
 - Creatine
 - Albumin
 - Globulin
 - CBC variables
 - Electrolytes



Workshop!

(Kidney Disease in Dogs)